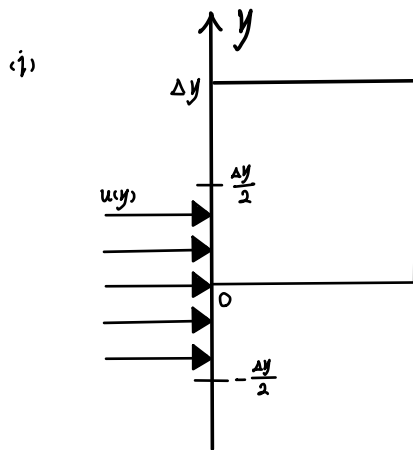


1. The x-component velocity, $u(y)$, normal to the face of a 2D control volume which has a centre at $y=0$ and whose height is Δy , so that $-\Delta y/2 < y < \Delta y/2$, varies as

$$u(y) = u_0 + a y + b y^2$$

(i) Draw a diagram of the problem.

(ii) Show that the average velocity, $\bar{U}$, over the face is $\bar{U} = u_0 + b (\Delta y)^2/12$



(ii)

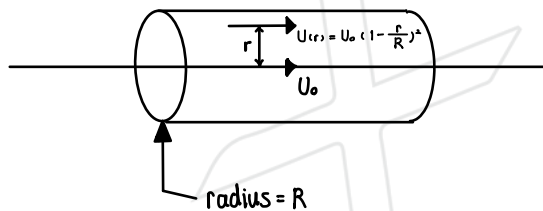
$$\begin{aligned} \bar{U} &= \frac{\int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} u(y) dy}{\Delta y} \\ &= \frac{1}{\Delta y} \left[u_0 y + \frac{1}{2} a y^2 + \frac{1}{3} b y^3 \right]_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} \\ &= \frac{1}{\Delta y} \left[u_0 (\Delta y) + \frac{1}{12} b (\Delta y)^3 \right] \\ &= u_0 + \frac{1}{12} b (\Delta y)^2 \end{aligned}$$

2. The velocity profile across a circular pipe of radius R is given by $U(r) = U_0(1 - (r/R)^2)$, where U_0 is the velocity at the centre and r is radial distance. This is a similar problem to question 3 but requires integration over a circular cross section.

To perform this integral we can use the method of integration of a *solid of revolution* (this will be covered in more detail in your mathematics lectures). In this approach we consider the volume under a surface $U(r)$ as the summation (or integral) of a series of circular areas (πr^2) of radius $r(U)$ multiplied by a thickness du and so the volume is

given by
$$\int_0^{U_0} U(r) dA = \int_0^{U_0} \pi r^2(U) du$$

- (i) Derive an expression for the mass flow rate in the pipe in terms of U_0 , r , R and ρ assuming steady flow. Recall that mass flow $Q = \rho \bar{U} A = \rho \int U dA$ where $\bar{U}$ is the average velocity, and A is the cross-sectional area.
- (ii) If the diameter of the pipe is 4cm and the velocity at the centre is 1m/s, calculate the Reynolds number Re , assuming the fluid in the pipe is air, defining Re as $\rho \bar{U} D / \mu$, where $\bar{U}$ is the average velocity of the air in the pipe.
(Ans $\rho \pi U_0 R^2 / 2$, 1308 using $\rho = 1.19 \text{ kg/m}^3$ and $\mu = 1.82 \times 10^{-5} \text{ Kg m/s}$)



$$(i) \quad Q = \rho \bar{U} A = \rho \frac{U(r) + U_c(r)}{2} (\pi R^2) = \frac{1}{2} (\rho U_0 \pi R^2)$$

$$(ii) \quad R = \frac{1}{2} \times 4 \text{ cm} = 2 \text{ cm}$$

$$\bar{U} = \frac{U_0}{2} = \frac{1}{2} \text{ m/s}$$

$$\therefore Re = \frac{\rho \bar{U} D}{\mu} = \frac{1.19 \text{ kg/m}^3 \times 0.5 \text{ m/s} \times 4 \text{ cm}}{1.82 \times 10^{-5} \text{ kg m/s}}$$

$$= 1307.69$$

$$= 1310$$

3. What is the vorticity of the following flows?

$$u = x, v = y$$

$$u = 1 - y^2, v = 0$$

(Ans 0, $2y$)

$$\text{For } u = x, v = y$$

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} = 1 - 1 = 0$$

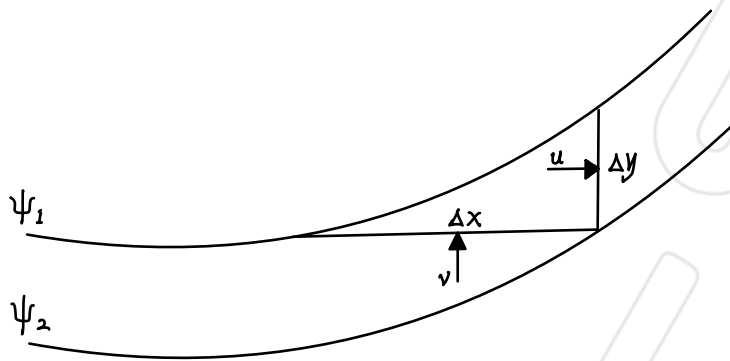
$$\text{For } u = 1 - y^2, v = 0$$

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 - (1 - y^2) dy = 0 - (-2y) = 2y$$

4. By considering the flow rate between two streamlines and recalling that the streamfunction is constant along a streamline and is related to the flow rate between the streamlines derive the following differential expressions:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

(Hint: Start with a diagram and also review lecture 2-4)



$$\psi_1 - \psi_2 = \int u \, dy = u \Delta y \quad \therefore \Delta \psi = u \Delta y$$

$$\psi_2 - \psi_1 = \int v \, dx = v \Delta x \quad \therefore -\Delta \psi = v \Delta x$$

$$\therefore \text{Let } \psi, x, y \longrightarrow 0$$

$$\delta \psi = u \delta y \quad \therefore u = \frac{\delta \psi}{\delta y}$$

$$-\delta \psi = v \delta x \quad \therefore v = -\frac{\delta \psi}{\delta x}$$